

Literature Status: Removing Compact Generation from the LCA Balian–Low Theorem

Run fa_banach_001 — lane 13

22 July 2026

Status: literature already answered. The open problem in the introduction of arXiv:1905.06827 asks whether the compact-generation hypothesis can be removed from the Kaniuth–Kutyniok zero theorem. The separate later paper arXiv:2001.07080 explicitly removes it in Theorem 1.7, with the stronger characterization in Theorem 5.8.

Original question

U. Enstad, *The Balian–Low theorem for locally compact abelian groups and vector bundles*, arXiv:1905.06827 [1], recalls that Kaniuth and Kutyniok proved the relevant Balian–Low statement for every lattice in a compactly generated, second-countable locally compact abelian group whose identity component is noncompact. The introduction then asks, on arXiv PDF page 3:

Source question. *Can the hypothesis that G is compactly generated be dropped from the Kaniuth–Kutyniok theorem?*

In the equivalent Zak-transform formulation used in the paper, the desired conclusion is that for every lattice $\Lambda \subseteq G$ and every $f \in L^2(G)$ for which $Z_{G,\Lambda}f$ is continuous, the Zak transform must have a zero.

Separate later answer

U. Enstad, M. S. Jakobsen, F. Luef, and T. Omland, *Deformations and Balian–Low theorems for Gabor frames on the adeles*, arXiv:2001.07080 [2], explicitly says on arXiv PDF page 4 that the compact-generation assumption can be removed and states:

Later answer (Theorem 1.7 / Theorem 5.8). *Let G be a locally compact abelian group with noncompact identity component. For every lattice Λ in G and every $f \in L^2(G)$ for which $Z_{G,\Lambda}f$ is continuous, $Z_{G,\Lambda}f$ has a zero.*

This is Theorem 1.7 in the introduction. The full statement appears as Theorem 5.8 on arXiv PDF page 24: for any locally compact abelian group G and lattice Λ , noncompactness of the identity component is equivalent to every continuous Λ -quasiperiodic function on $G \times \widehat{G}$ having a zero. The Zak-transform conclusion follows in particular.

The supporting paper also cites and uses Theorem 5.5 of arXiv:1905.06827 as its Proposition 5.7, and explicitly describes Theorem 5.8 as removing compact generatedness. Thus the authors knew they were answering precisely this problem; no new mathematical implication is claimed here.

Scope and search status

The later theorem completely answers the source question and in fact states the result for arbitrary locally compact abelian groups in its formulation, whereas the source question was posed in the second-countable setting. No part of that compact-generation-removal question remains open.

A bounded status check on 22 July 2026 searched the run’s registry and solution/attempt indexes, the local arXiv source corpus for arXiv:1905.06827, the exact phrase “the hypothesis that G is compactly generated can be dropped”, and primary-source searches for Balian–Low/Zak-transform results. The decisive match was arXiv:2001.07080; its local source confirms Theorems 1.7 and 5.8 and its explicit reliance on arXiv:1905.06827.

References

- [1] U. Enstad, *The Balian–Low theorem for locally compact abelian groups and vector bundles*, arXiv:1905.06827 (2019).
- [2] U. Enstad, M. S. Jakobsen, F. Luef, and T. Omland, *Deformations and Balian–Low theorems for Gabor frames on the adeles*, arXiv:2001.07080 (2020; revised 2022).